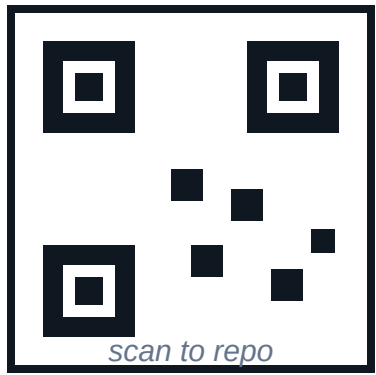


Measuring Neural-Network Robustness Under Adversarial Perturbation



Experimental Evaluation of Model Behavior Under Gradient-Based Attacks

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Security Engineering • AI / Agent Security • Cloud Security

MIT • Python 3.12 • HEAD 0fadaa2 • verified 2026-09-26

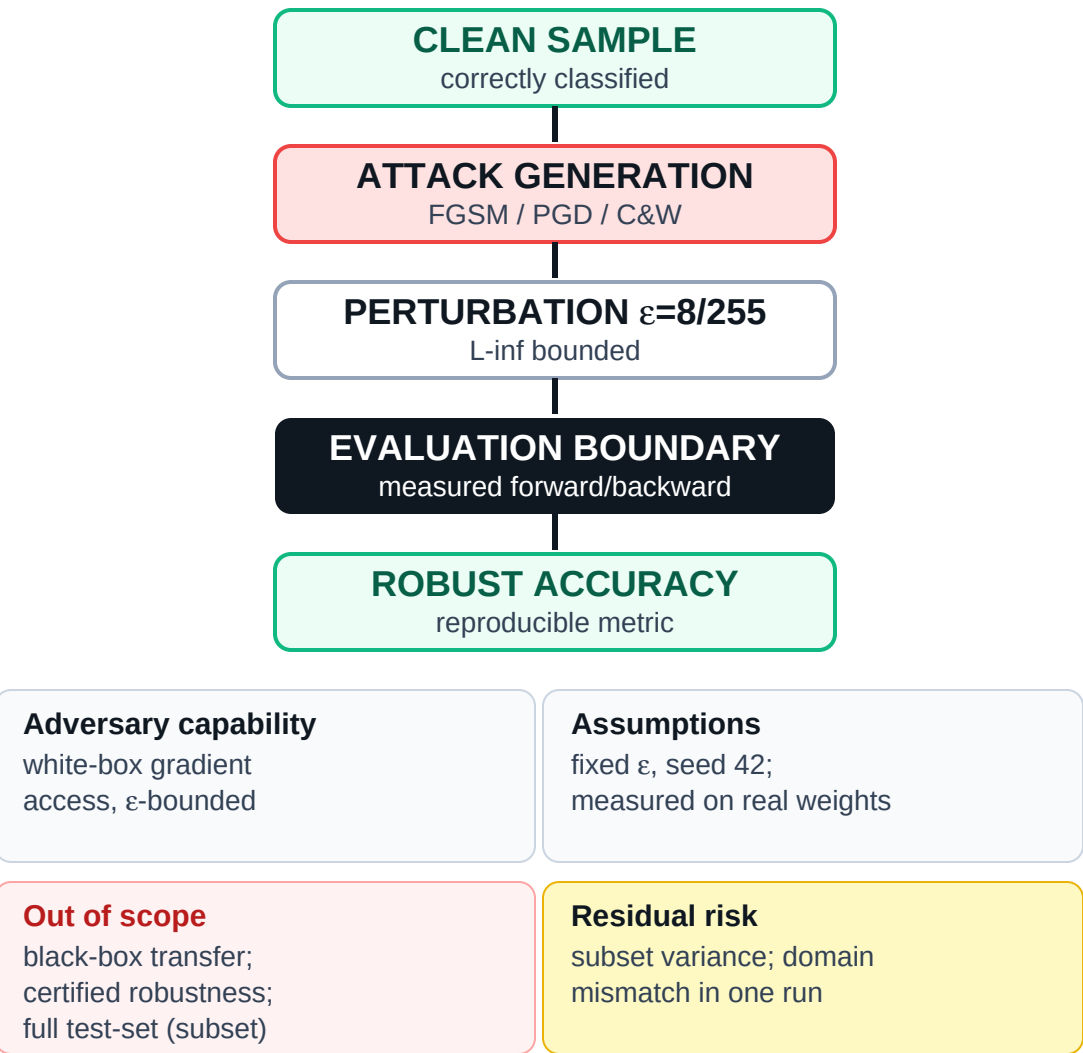
RESEARCH QUESTION

How does a model's accuracy collapse under FGSM/PGD/C&W attacks — and are the reported numbers actually measured, not projected from literature?

01 • THE FAILURE MODE

Small L-infinity perturbations flip confident model predictions. Reporting robustness is easy to fake: a single-step FGSM can look 'robust' due to gradient masking, and literature values can be pasted in as if measured. This lab separates REAL measured runs from clearly-labeled literature projections.

02 • THREAT MODEL



03 • OBJECTIVE

Measure real robustness collapse under gradient attacks and keep measured results strictly separate from literature projections.

- 01 – FGSM / PGD / C&W attack impls
- 02 – Measured runs on real weights
- 03 – Robust-accuracy metrics + timing
- 04 – Label projected numbers as projected

HERO • ADVERSARIAL EVALUATION LOOP

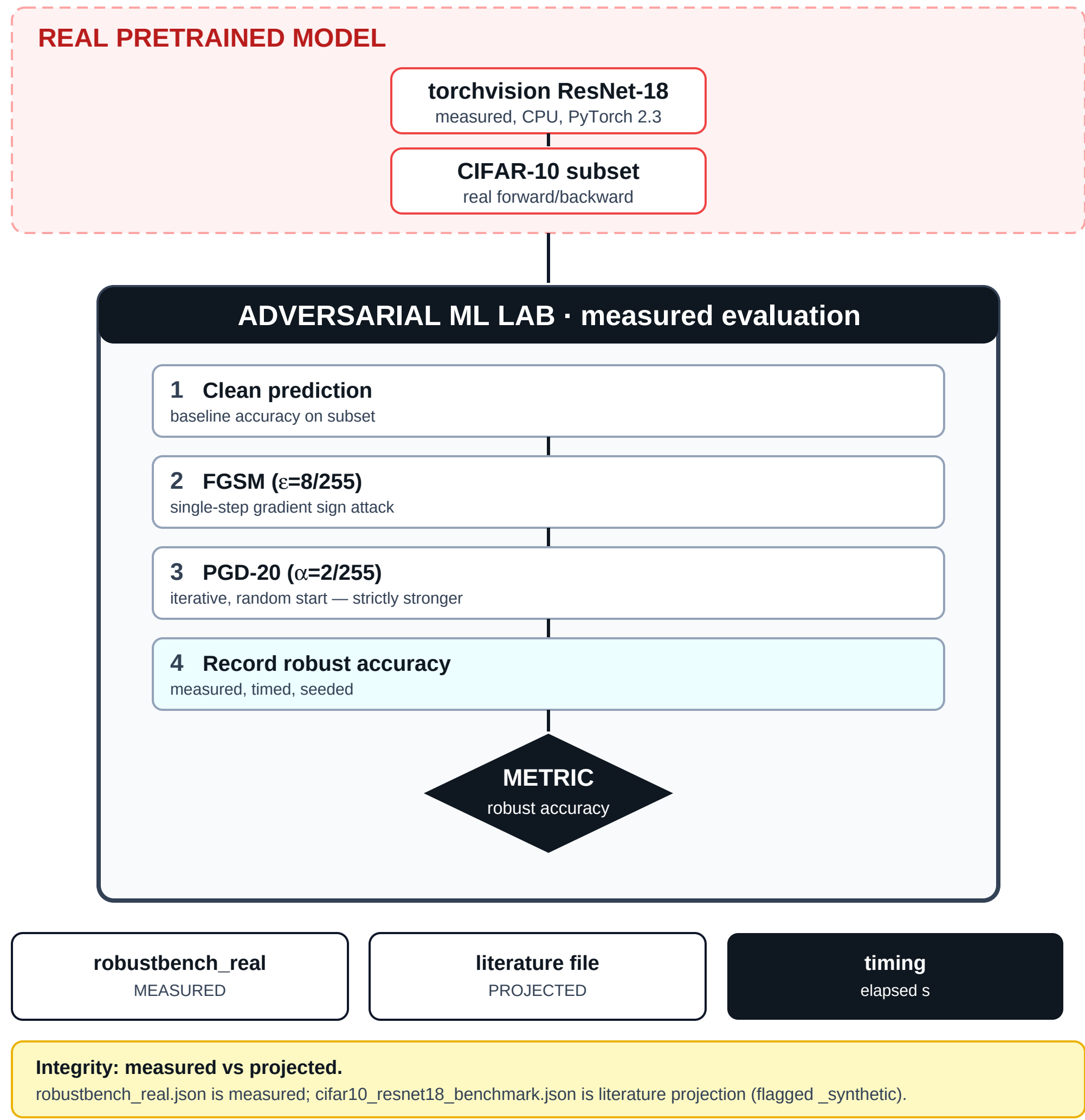
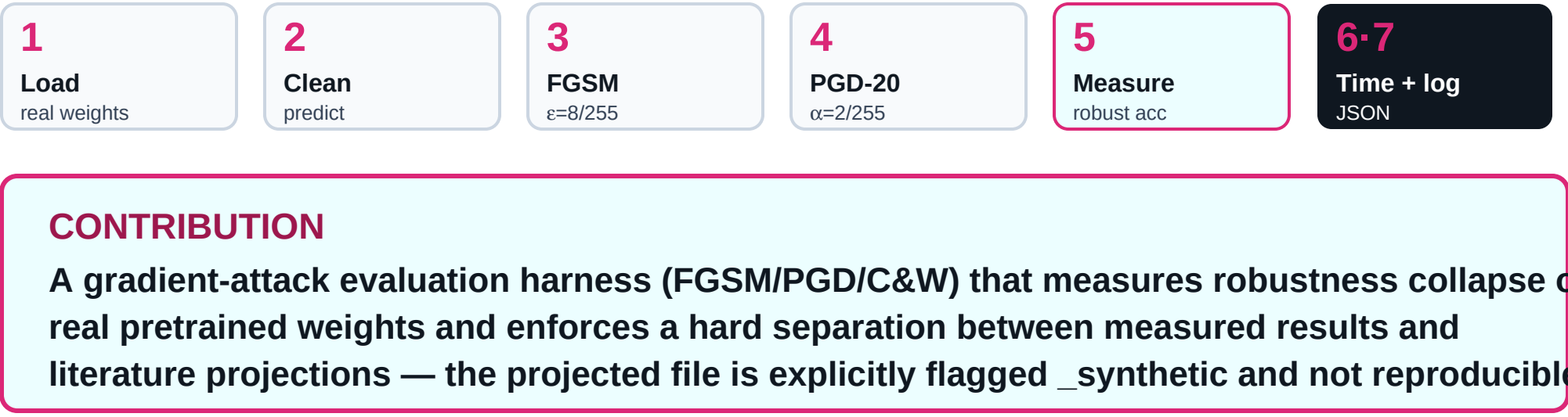


Figure 1. Measured adversarial loop on real pretrained weights. Projected literature values are stored separately and NOT shown as measured.

METHOD • ATTACK EVALUATION



RESULTS • MEASURED ATTACK SUCCESS

100%

PGD-20 SUCCESS

100%

FGSM SUCCESS

Measured on real weights, CIFAR-10 subset (robustbench_real.json). On correctly-classified inputs; PGD-20 stronger than FGSM (no masking).

Measured config
Attack budget ϵ 8/255
PGD steps 20
Framework torch 2.3

ADVERSARIAL ACCURACY (measured subset)
Clean (subset) 5%
FGSM adv acc 0%
PGD-20 adv acc 0%

Real measured run; clean acc low due to ImageNet→CIFAR domain mismatch on the subset

LITERATURE PROJECTION (labeled)
cifar10_resnet18_benchmark.json is a LITERATURE PROJECTION (Madry 2018): undefended PGD ~0%, Madry-AT ~45%. NOT measured here — shown as context.

NOT ESTABLISHED BY THIS REPO
State-of-the-art robustness. Certified guarantees. Full-test-set numbers. Black-box transferability.

SECURITY BOUNDARY
PROTECTS / DETECTS

- measures FGSM/PGD/C&W
- real weights, real data
- gradient-masking check
- seeded + timed
- measured vs projected

DOES NOT ESTABLISH

- does not defend the model
- no certified bound
- no SOTA claim
- subset, not full set

MEASURED ≠ PROJECTED

LIMITATIONS

- 1. Measured run uses a small subset, not full test set.
- 2. Clean accuracy low (ImageNet→CIFAR domain mismatch).
- 3. Literature file is projection, not reproducible.
- 4. White-box only; no black-box/transfer attacks.
- 5. No certified-robustness claim.

FUTURE WORK

- Full 10k CIFAR-10 test-set measured run.
- CIFAR-trained model (remove domain mismatch).
- Madry adversarial-training measured baseline.
- AutoAttack ensemble evaluation.
- Certified-robustness comparison.

CONCLUSION

The lab measures real robustness collapse under gradient attacks and keeps measured results strictly apart from literature projections. On the tested subset, FGSM and PGD-20 reach 100% success. It evaluates robustness; it does not certify it.

REPRODUCE
python benchmark/robustbench_baseline.py
pytest tests/